

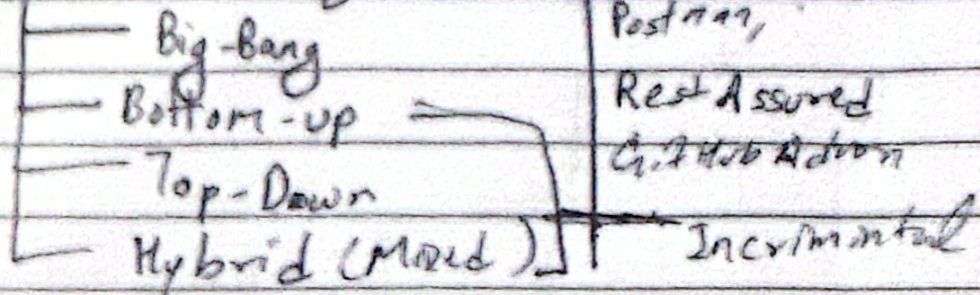
* Is Null and Is Not Null
 ≠ use java null di mana...

Day 3 Week 2

Testing levels

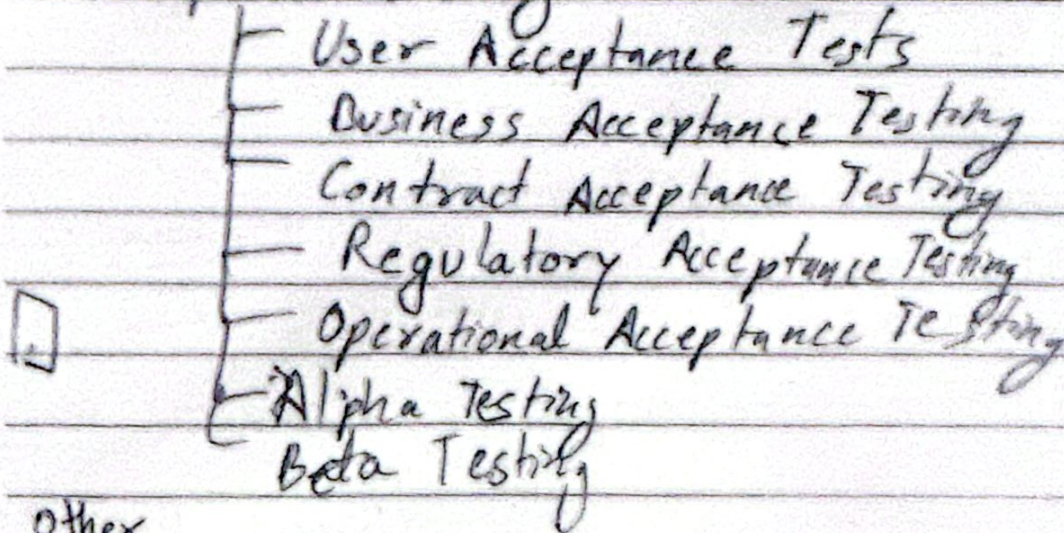
Unit Testing

Integration Testing



System Testing

Acceptance Testing



Other

Path Testing

Loop Testing

Simple Loop

- Nested Loop
- Concatenated Loop
- Unstructured Loop

Mutation Testing

- Value mutations
- Decision Mutations
- Statement Mutations.

Penetration Testing

Stages - Planning & Reconnaissance

- Scanning
- Gaining access
- Maintaining access
- Analysis and WAF configuration.

Dynamic

Structure Based

(White box testing)

- Decision Coverage
- Statement Coverage
- Condition Coverage
- Multiple Condition
- Base path Testing
- Loop testing

Experience Based

- Error Guessing
- Exploratory Testing

Specification Based

(Black-box testing)

- Equivalence Partitioning
- Boundary Value Analysis
- Decision Tables
- Use Case Testing
- State Transition

Stubs

Drivers

Testing Pyramids

Formula to calculate

Branch coverage / Decision
statement coverage
condition coverage
multiple conditions

Penetration Testing Methods

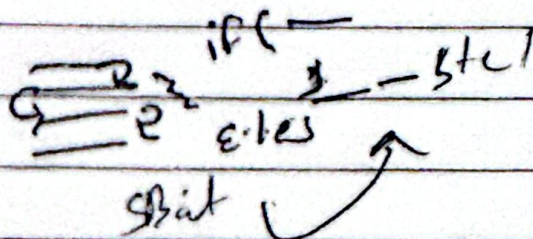
External Testing

Internal Testing

Blind Testing

Double-Blind Testing

Targeted Testing.



Statement Coverage

It measures the percentage of executable statements that have been tested in a program
Eg:-

Read A $\rightarrow$ statement 1

Read B $\rightarrow$ st 2

if $A > B \rightarrow$ st 3

Print "A is bigger" $\rightarrow$ st 4

else $\rightarrow$ st 5

Print "B is bigger" $\rightarrow$ st 6

endif. st $\rightarrow$ 7

Case 1: $A = 5, B = 3$

No. of statements executed = 5

Total no. of statements = 7

$$\text{statement coverage} = \left(\frac{5}{7} \right) \times 100\%$$

$$= 71\%$$

Case 2: $A = 3, B = 7$

No. of statements executed = 6

Total no. of statements = 7

$$\text{statement coverage} = \left(\frac{6}{7} \right) \times 100\%$$

$$= 85.7\%$$

Branch Testing

It checks whether every possible outcome of decision has been tested.

Eg:-

```
if (age >= 18)           // Decision 1
    print ("Adult");
else
    print ("Minor");
```

True
False
2 branches

```
if (age >= 65)
    print ("Senior");
else
    print ("Not senior");
```

2 branches

```
if (has ID)
    print ("ID verified");
else
    print ("No ID");
```

2 branches

Case 1: age = 70

has ID = true

No. of total branch = 6 branch

No. of executable branch = 3 branch

Case 2: age = 15

has ID = false

No. of total branch = 6 branches

No. of executable branch = 3 branch



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Memo No. _____

Date / /

$$\text{Total branch coverage} = \left(\frac{\text{total tested branches}}{\text{total branches}} \right) \times 100\%$$

$$= \left(\frac{6}{6} \right) \times 100\%$$

for all 2 cases. $\approx 100\%$